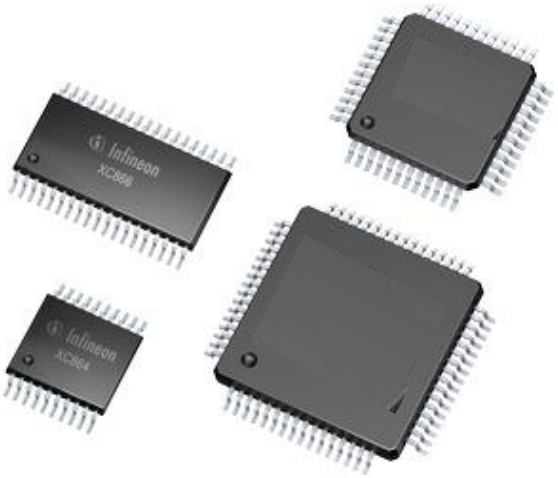
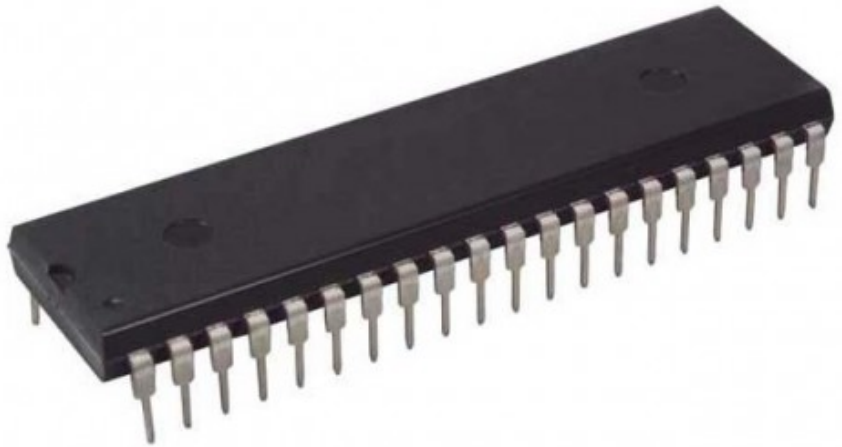
# 8051 Microcontroller

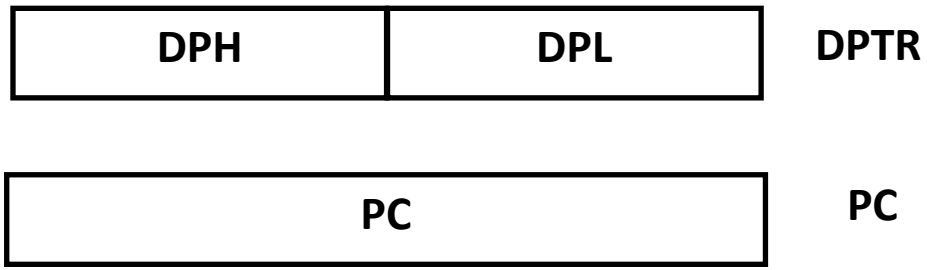


# Registers of 8051

| A |
|---|
| B |
| R0 |
| R1 |
| R2 |
| R3 |
| R4 |
| R5 |
| R6 |
| R7 |

**Some 8-bit Registers of the 8051**




**Some 8051 16-bit Register**

**Accumulator Register (A)**

- 8 bit register including add, sub, mul, div and boolean manipulation
- Used for data transfer operation from and to external memory

**Register B**

- 8 bit register is used with A reg. for manipulation for multiplication and division
- Has no location in external memory that are address by pc.

**Program Counter (16 bit):**

- Program counter used to point the memory location where the next instruction to be fetched.

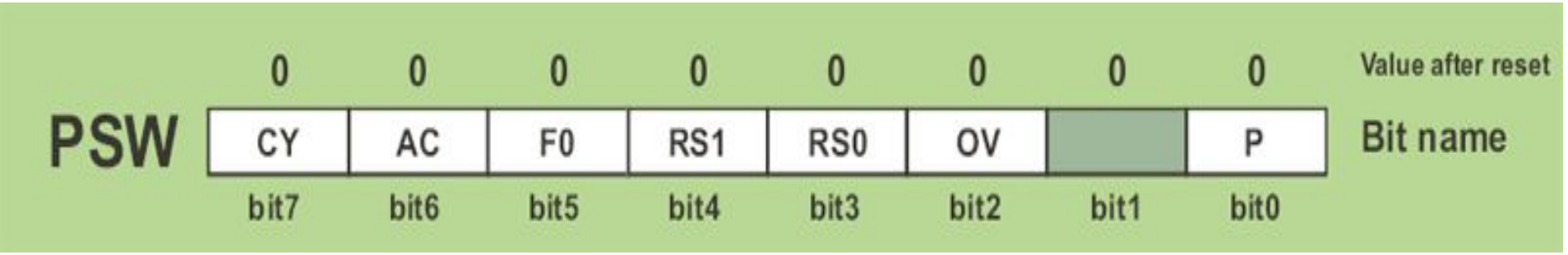
**Stack Pointer SP (8 bit):**

- When CPU is powered sp points to 07H by its own.
- Program counter is automatically incremented after the execution of instruction byte is fetched.

**Data pointer(16 bit)**

- Made up of two 8bit registers DPH and DPL
- DPTR is under the control of program instruction and can be specified by 16 bit name, DPTR or any other individual byte name, DPH,DPL

# Program Status Word (PSW) Register




**Parity flag**

- If A reg. contains an odd number of 1's then P=1
- If A reg. contains an even number of 1's then P=0

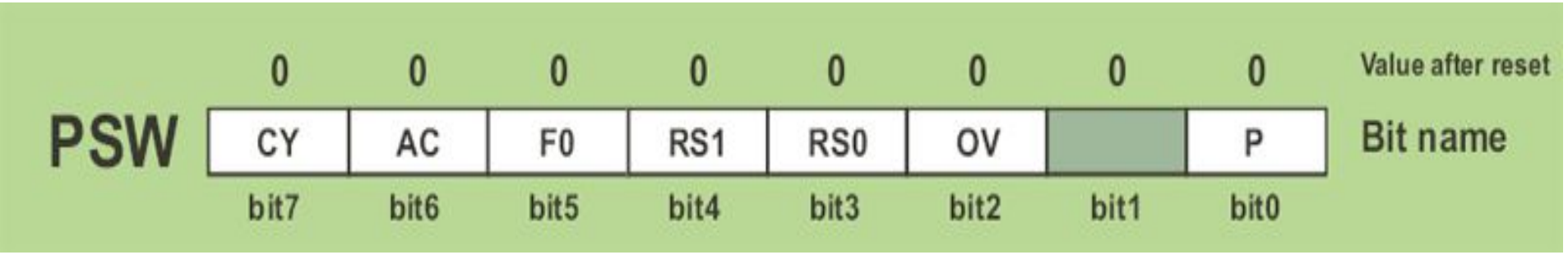
**Overflow flag ov**

- occurs when the result of an arithmetical operation is larger than 255 and cannot be stored in one register. Overflow condition causes the OV bit to be set (1). Otherwise, it will be cleared (0).

**F0 flag**

- Available to user for general purpose

## Program Status Word (PSW) Register




**Auxiliary carry flag AC**

- If there is a carry from D3 to D4 during ADD, SUB operation, this bit is set

**Carry flag CY**

- This flag is set whenever there is carry out from D7 bit.

**RS1 and RS0**

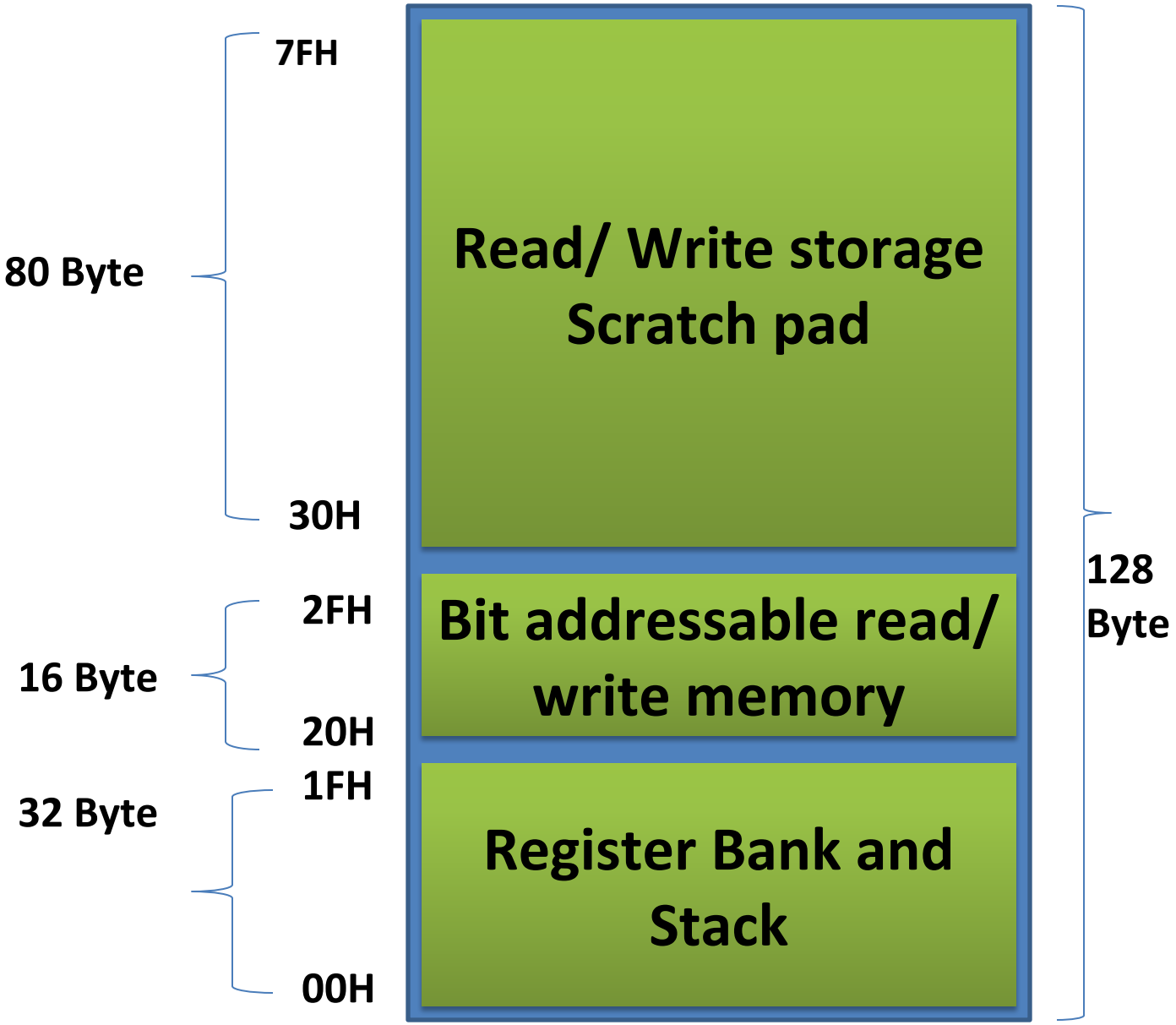
- Register bank selector

| RS1 | RS0 | Register Bank |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 2 |
| 1 | 1 | 3 |

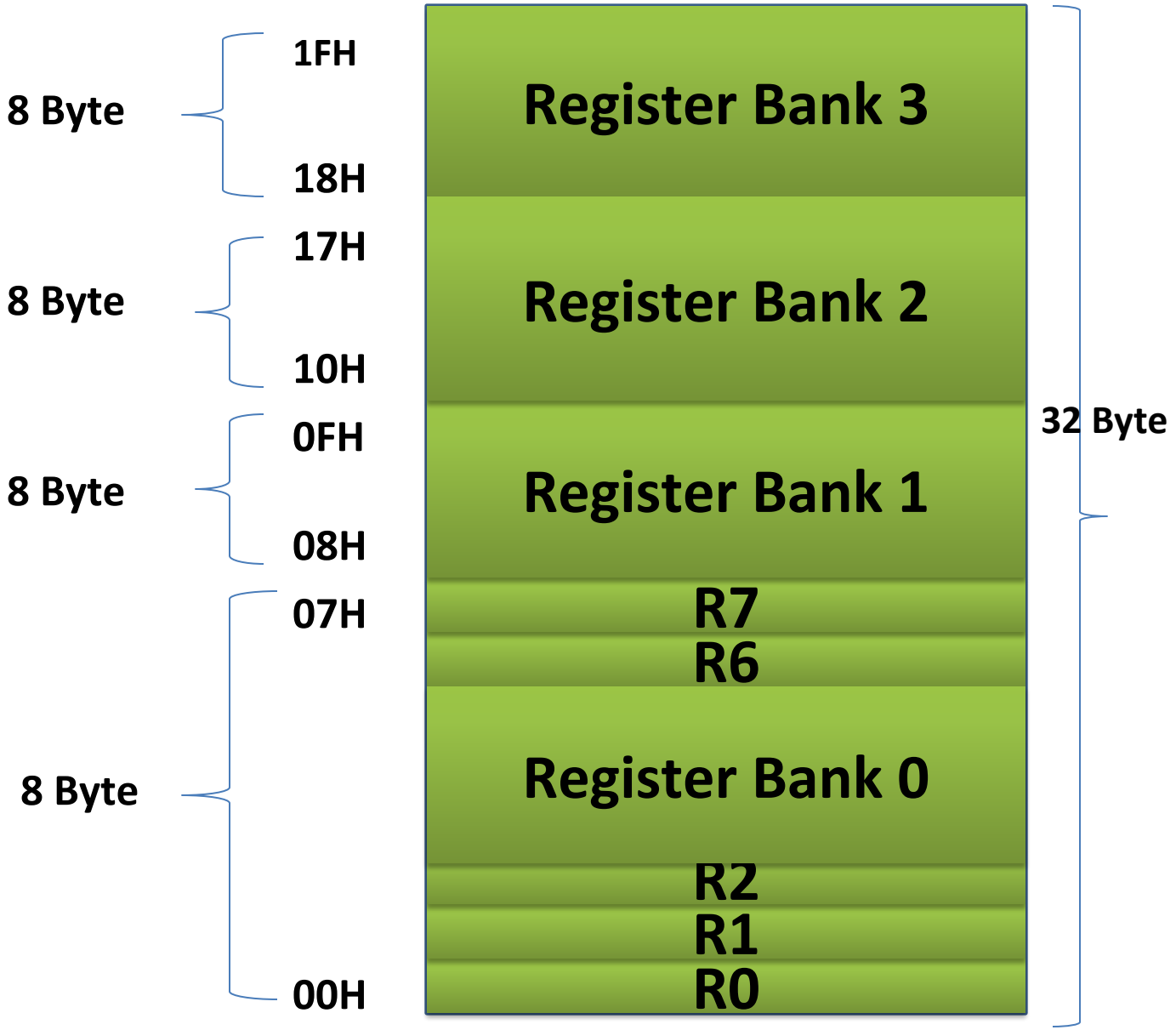
RAM=128 byte
Range=00 to 127
Range=00H to 7FH

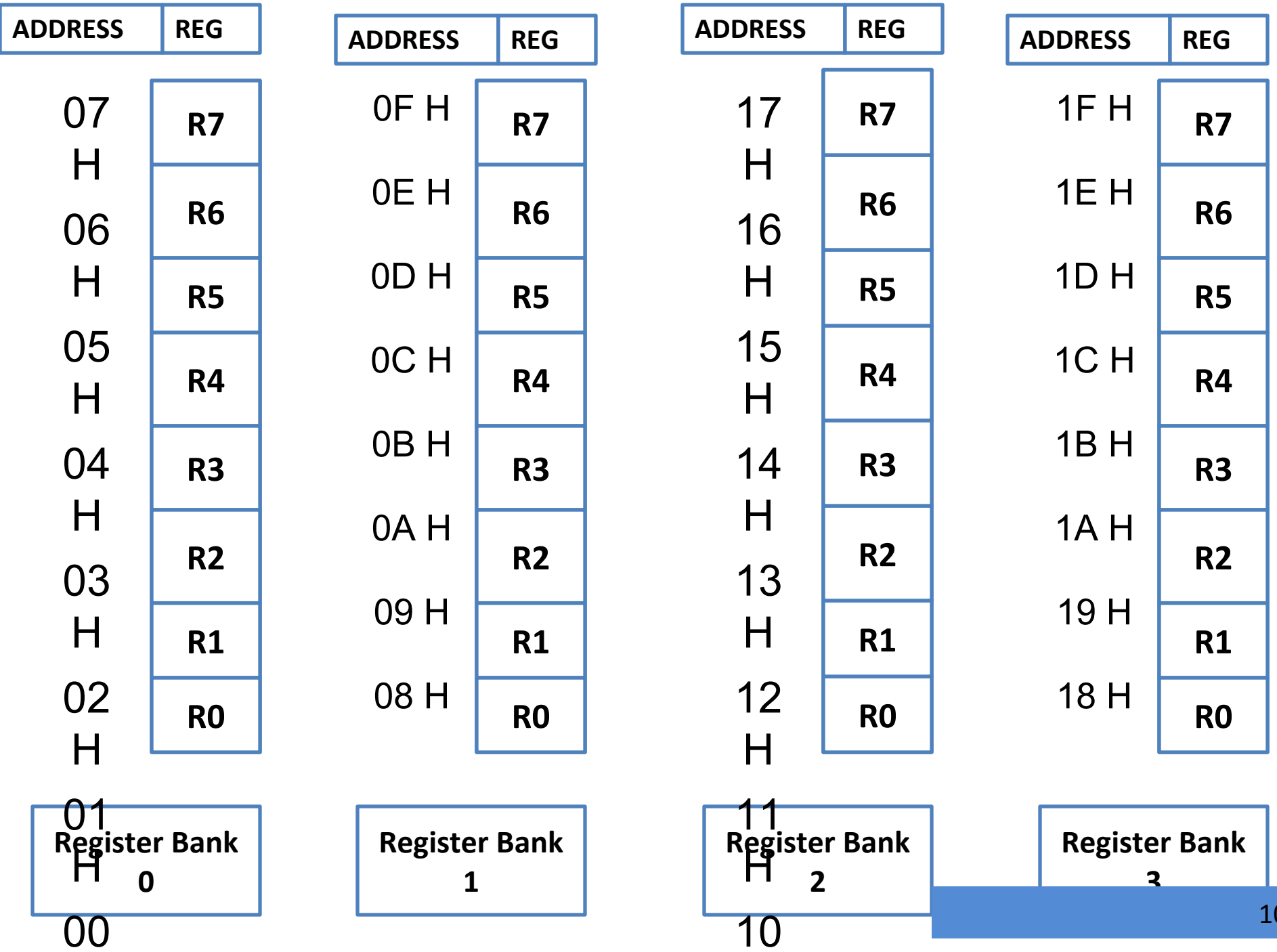
7F=127
1F=31

## 8051 Memory Organization

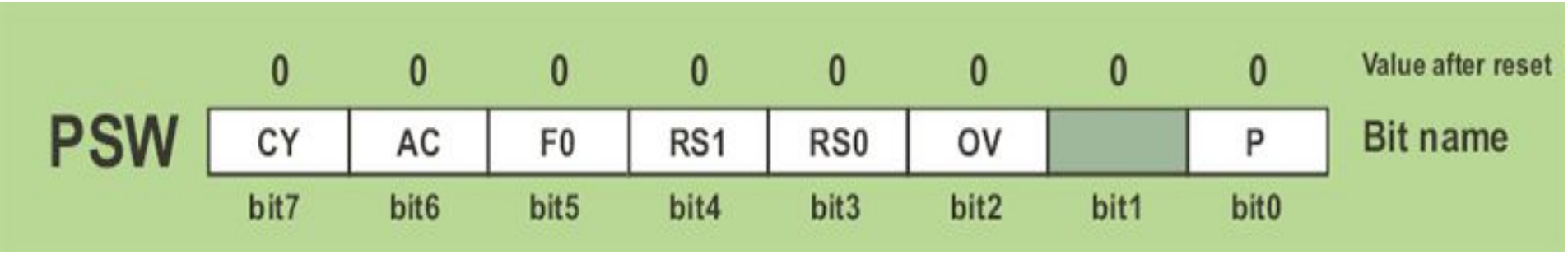


Register Bank and Stack




ADDRESS
REG
07 H
R7
06 H
R6
05 H
R5
04 H
R4
03 H
R3
02 H
R2
01 H
R1
00
R0
Register Bank 0
0F H
R7
0E H
R6
0D H
R5
0C H
R4
0B H
R3
0A H
R2
09 H
R1
08 H
R0
Register Bank 1
17 H
R7
16 H
R6
15 H
R5
14 H
R4
13 H
R3
12 H
R2
11 H
R1
10
R0
Register Bank 2
1F H
R7
1E H
R6
1D H
R5
1C H
R4
1B H
R3
1A H
R2
19 H
R1
18 H
R0
Register Bank 3

# How to switch register bank?




| RS1 | RS0 | Register Bank |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 2 |
| 1 | 1 | 3 |

- **Default register bank= Register Bank 0**

**MOV A,R0H ;** move data from R0 into reg. A

**MOV A, 00H;** move data from memory location 00h(R0) into reg. A

**Que. To move data from reg. R0 of register bank 2**

**SETB PSW.4**
**CLR PSW.3**
**MOV A, R0H**

} **MOV A,10H**

| RS1 PSW.4 | RS0 PSW.3 | Register Bank |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 2 |
| 1 | 1 | 3 |

# Stack in 8051

- Stack is a section of RAM used by CPU to store information temporary. This information could be data or an address

- The register used to access stack is called SP(stack pointer) register.

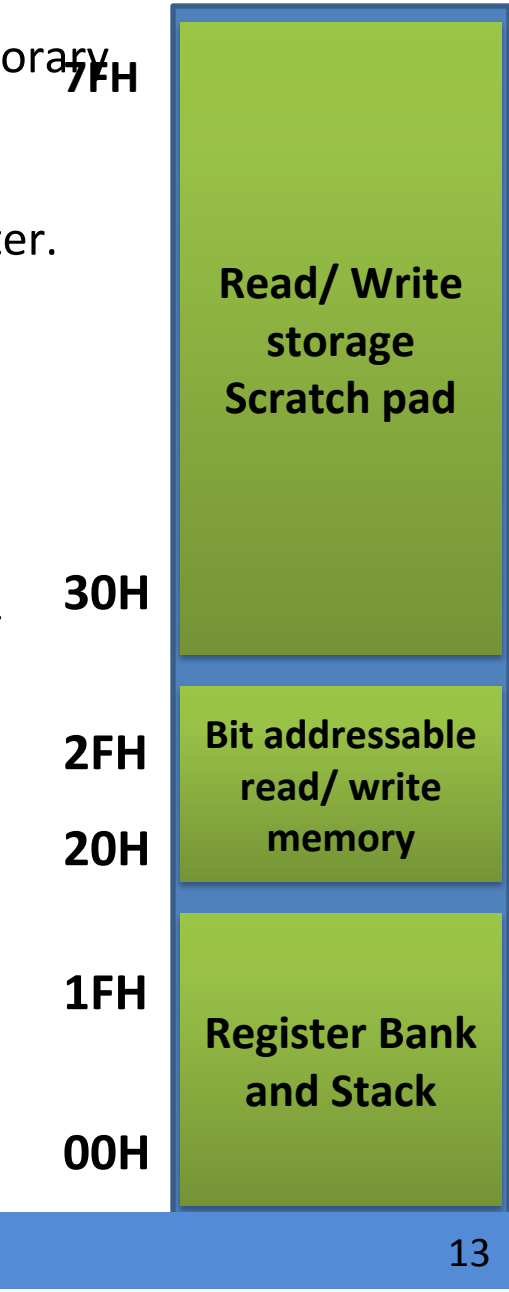
| 8086 | 8051 |
|---|---|
| SP=16 bit | SP=8 bit |

- Default value in SP reg. is 07H
  i.e RAM location 08H is the first location used for stack by 8051

- Register Bank 1= Default Stack memory

Location 08H to 1FH in 8051 RAM can be used for stack.

If in given program we need more than 24 bytes ( 08h to 1Fh) of stack, we can change SP to point RAM location 30 – 7FH using instruction
MOV SP,#RAM address

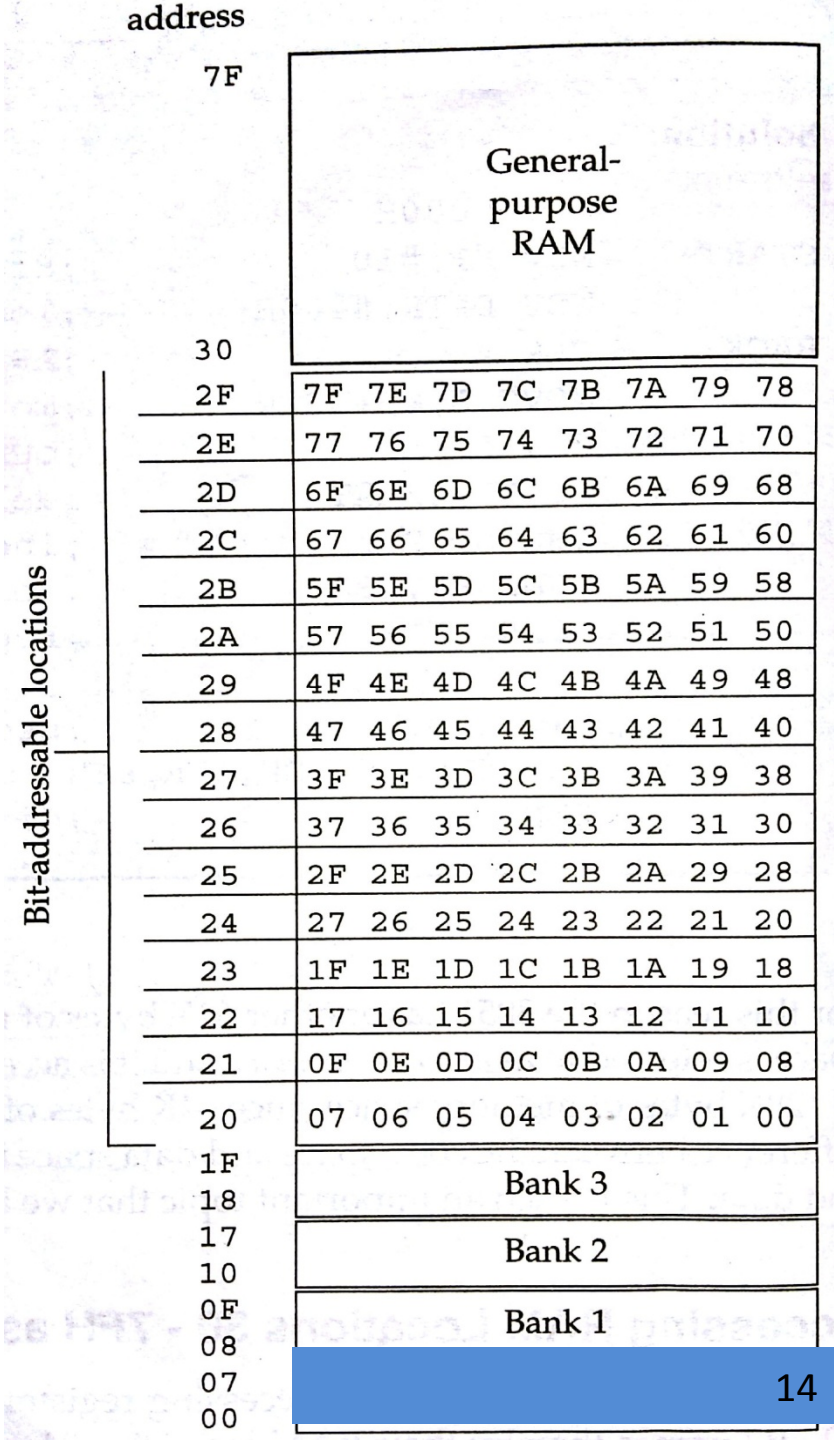


# Bit addressable read/ write memory

**16 byte**
**Range= 20H to 2FH**

**16 x 8 = 128 bits**

| Byte addressable | Bit addressable |
|---|---|
| Register=8 bit<br>Byte has address | Register=8 bit<br>Bit has address |
| Access registers in bytes | Access registers in bits |
| Ex. SP | Ex. I/O port reg. |



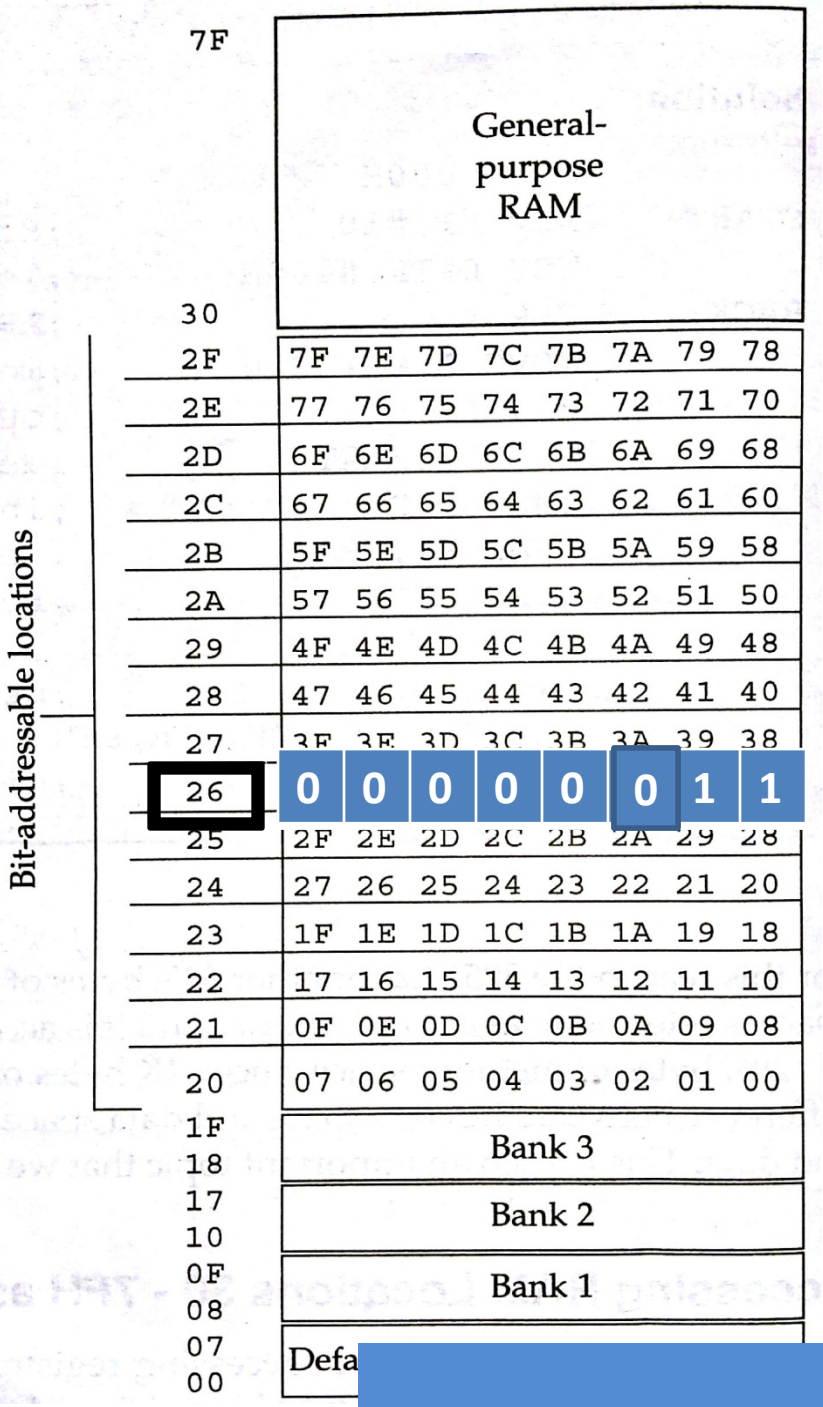
# Bit addressable read/ write memory

**MOV A, #03H;** copy byte 03h to reg. A

**MOV 26H,A;** copy data byte from reg. A to memory location 26H

**SETB 32H**

**CLR 32H**




16 byte

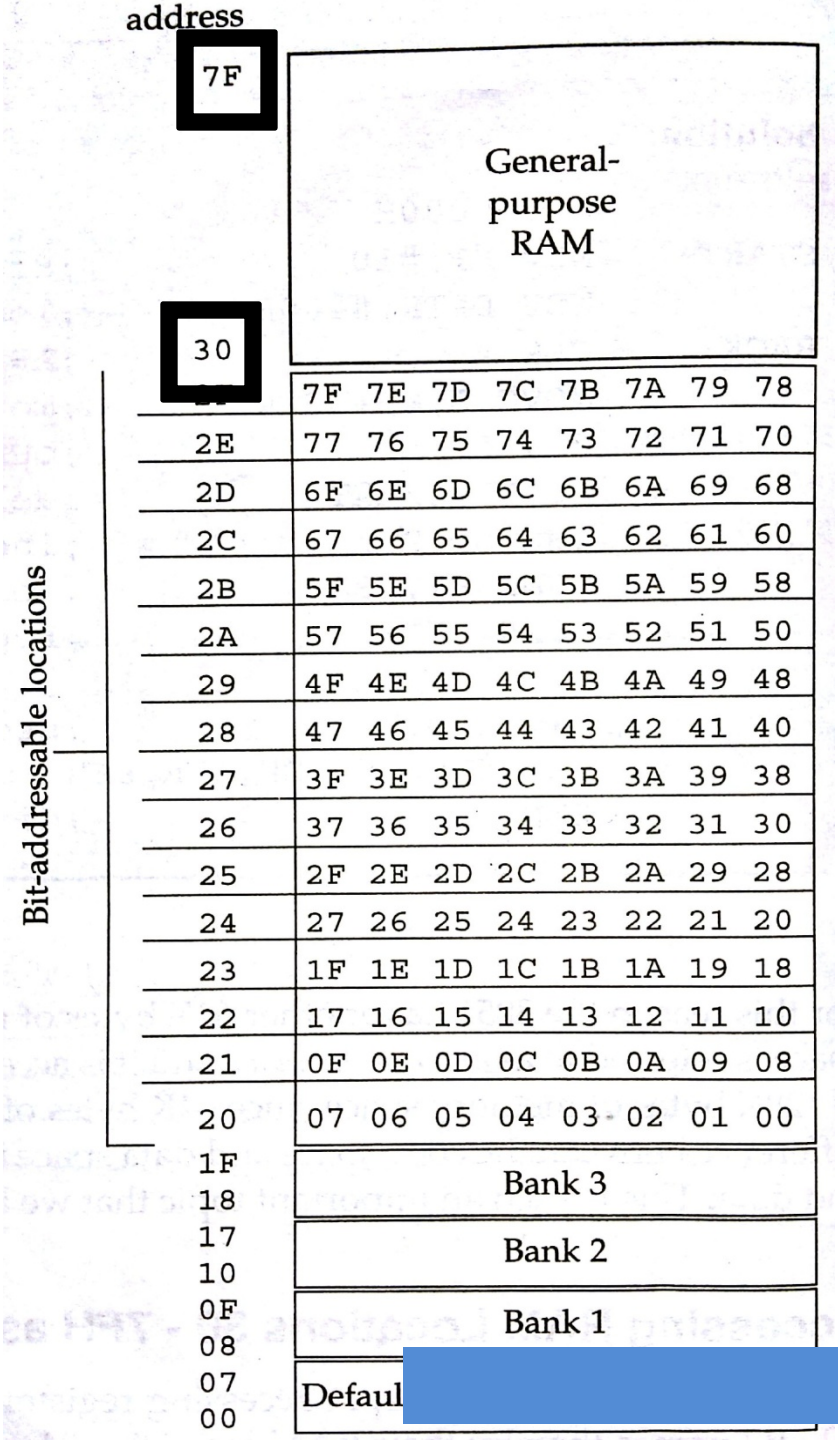
**MOV P0, #0FFH** BYTE addressable instruction

**SETB P0.0** BIT addressable instruction

| P0 | P0.7 | P0.6 | P0.5 | P0.4 | P0.3 | P0.2 | P0.1 | P0.0 |
|---|---|---|---|---|---|---|---|---|
| ADDRESS | 87H | 86H | 85H | 84H | 83H | 82H | 81H | 80H |

# Read/ Write storage Scratch pad

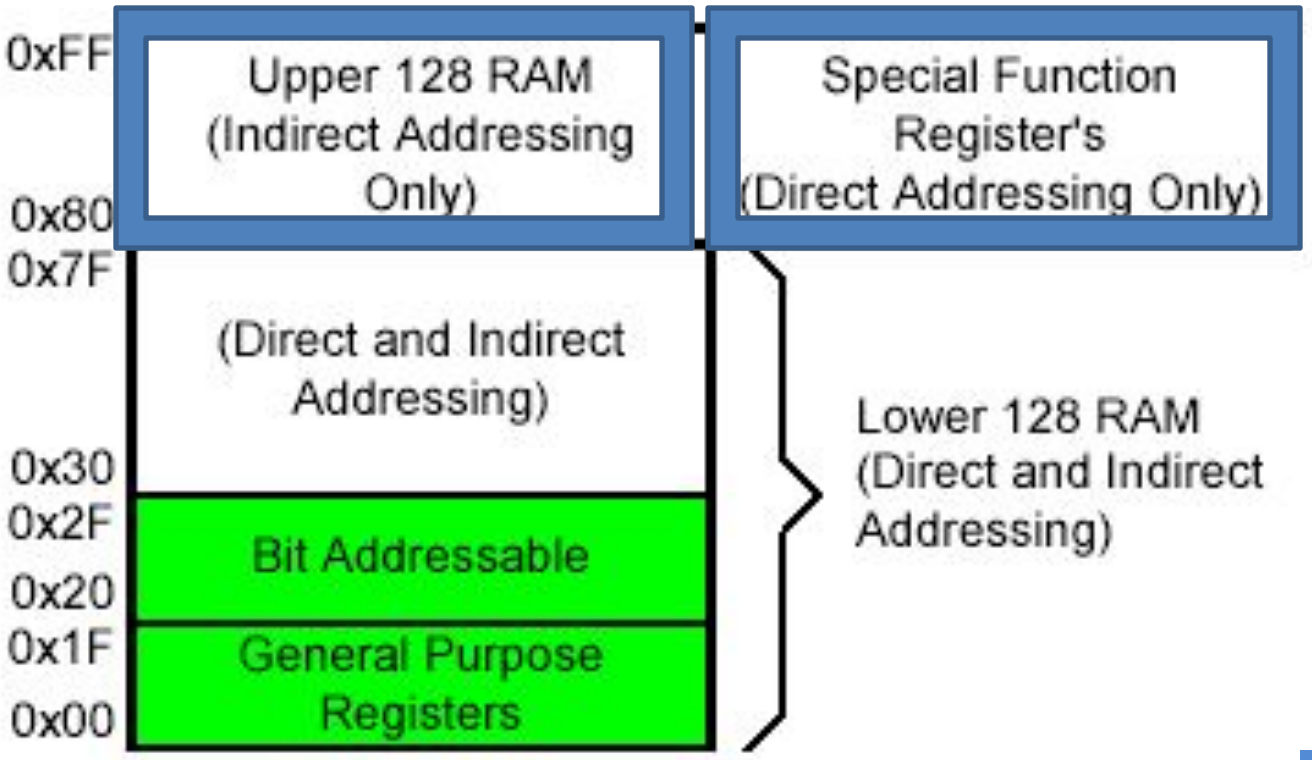
- Use for read and write storage
- General purpose storing of data
- If we need more registers we simply use RAM locations 30-7FH



## Extra 128 byte on chip RAM in 8052

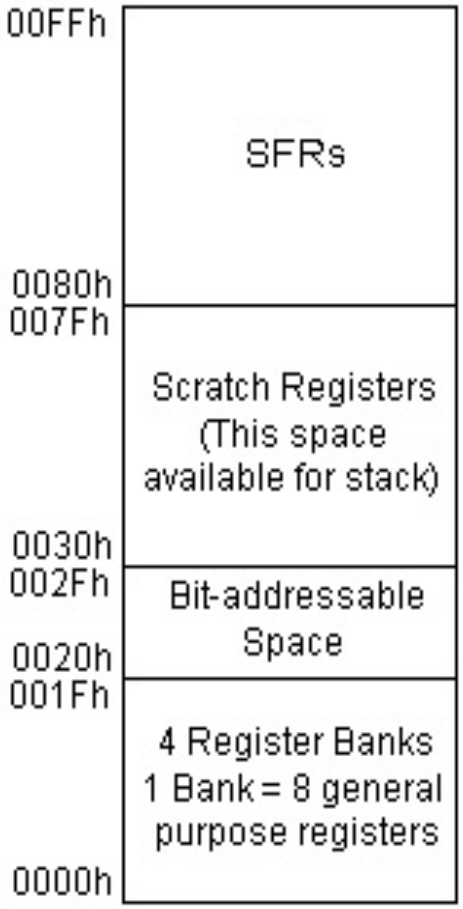
- 8052 is advanced version of 8051
- Another 128 bytes of on chip RAM with address 80-FFH are designated as special function register(SFR) / upper RAM

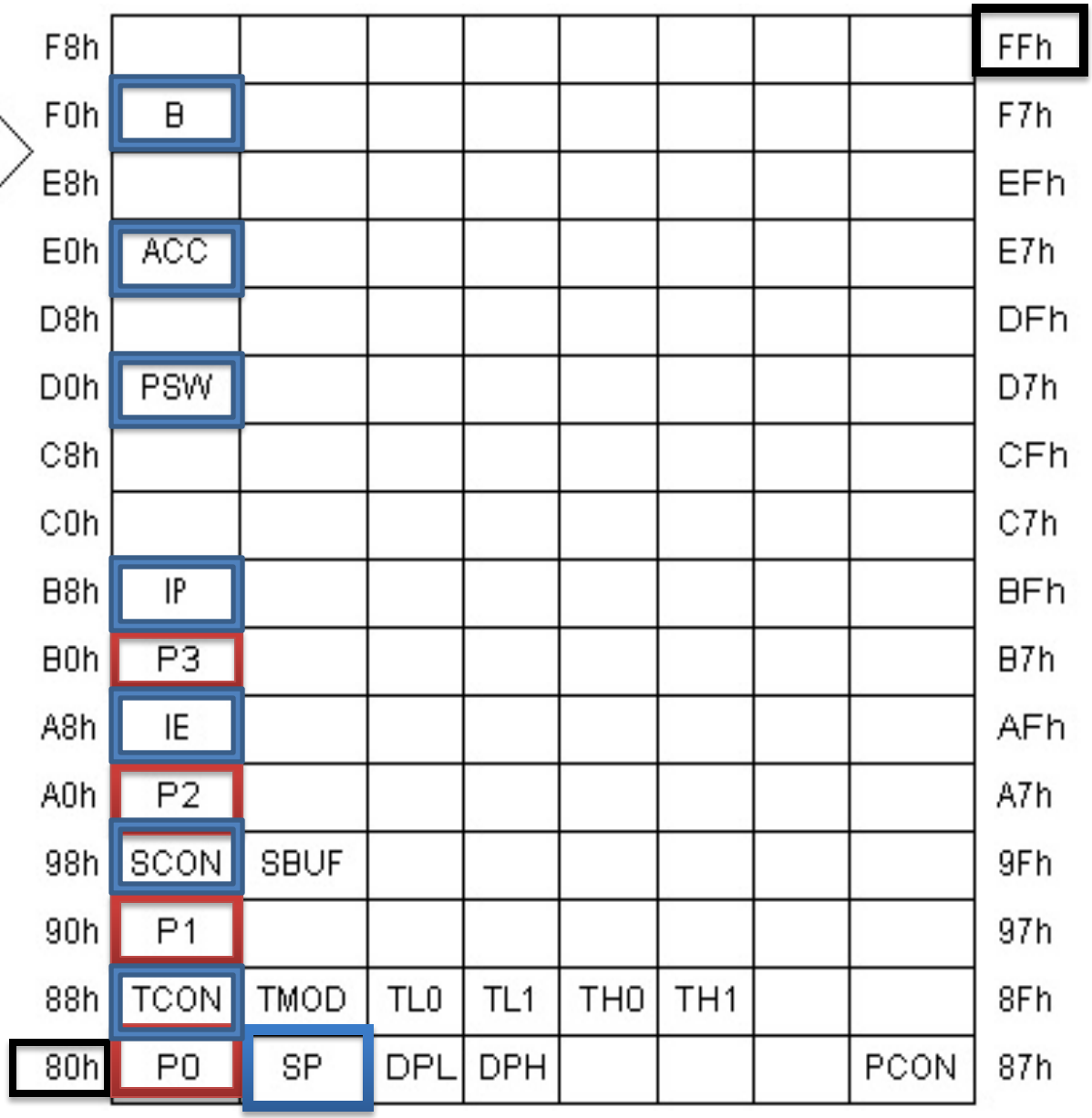
| 8051 | 8052 |
| --- | --- |
| 128 byte of RAM | 256 byte of RAM |




Two different physical memories, but they have same addresses

# SFR Special Function Register




**Bit addressable registers**

## ROM in 8051

- 4K bytes on chip ROM
- Use to store user code
- ROM=code memory
- Size of PC=16 bit
- Therefore maximum size of on chip ROM can be 0000 to FFFFFH i.e 64 Kbytes
- 4 K bytes of ROM memory has addresses from 0000H to 0FFFH.